RAJALAKSHMI ENGINEERING COLLEGE

V Y Aares 251801001-2025-AIDS V2

Started on	Thursday, 27 August 2026, 8:51 AM
State	Finished
Completed on	Thursday, 27 August 2026, 8:52 AM
Time taken	24 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Comparator function for qsort
int compare(const void *a, const void *b) {
    long long int_a = *(const long long *)a;
    long long int_b = *(const long long *)b;

    if (int_a < int_b) return -1;
    if (int_a > int_b) return 1;
    return 0;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    // Use long long to prevent integer overflow for large sums
    long long *arr = (long long *)malloc(n * sizeof(long long));
    for (int i = 0; i < n; i++) {
        scanf("%lld", &arr[i]);
    }

    // Step 1: Sort the array in O(n log n)
    qsort(arr, n, sizeof(long long), compare);

    // Step 2: Calculate maximum sum in O(n)
    long long maxSum = 0;
    for (int i = 0; i < n; i++) {
        maxSum += arr[i] * i;
    }

    // Step 3: Output the result
    printf("%lld\n", maxSum);

    free(arr);
    return 0;
}
```

	Input	Expected	Got	
✔	5 2 5 3 4 0	40	40	✔
✔	10 2 2 2 4 4 3 3 5 5 5	191	191	✔
✔	2 45 3	45	45	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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